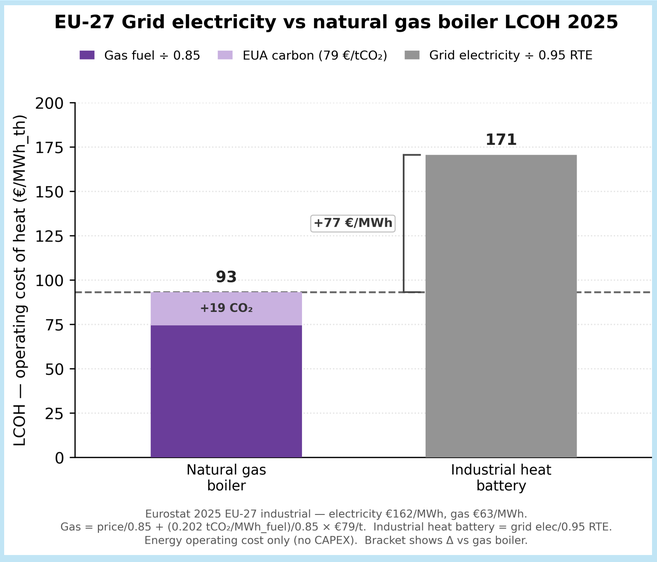
# Techno-economic Potential of Electrified Industrial Process Heat in European Manufacturing

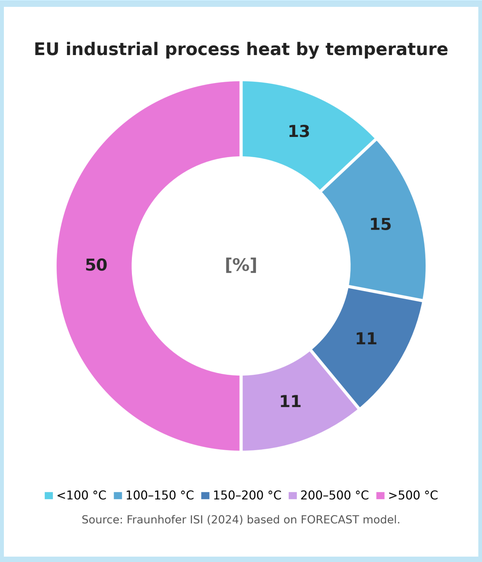
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## Introduction



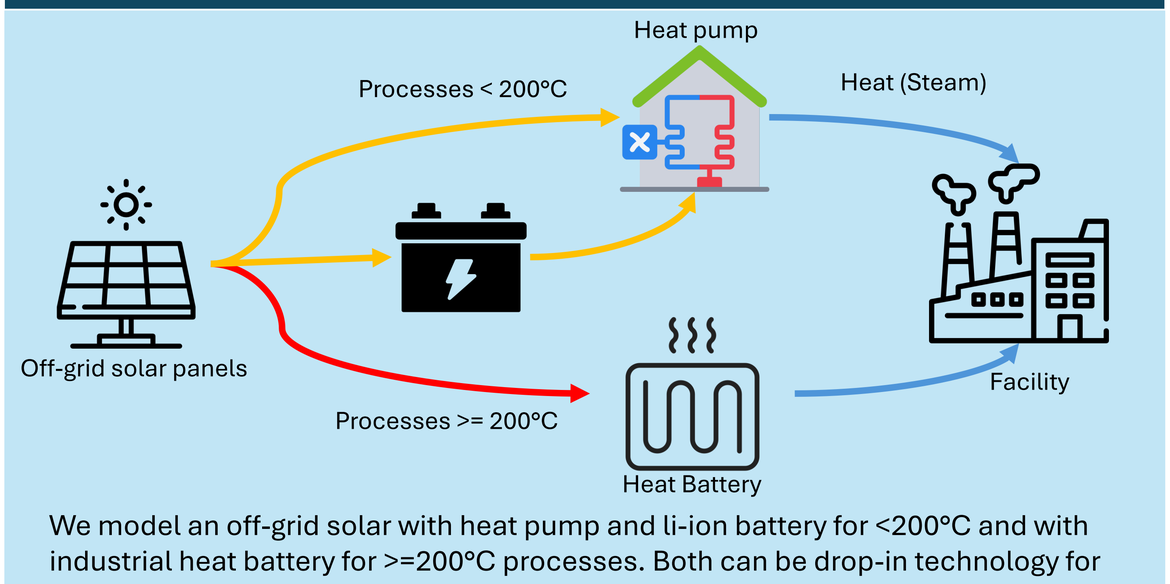

50% of EU industrial process heat demand is above 500°C.
Industrial heat batteries are the leading option to electrify it, yet grid power costs €77/MWh more than gas.

## Results

| | 5 km | 15 km |
|---|---|---|
| Served (max heat) | 58.8% (307 TWh) | 84.4% (441 TWh) |
| LCOH (max heat) | €55 | €62 |
| Served (max savings) | **53.4% (278 TWh)** | **73.9% (377 TWh)** |
| LCOH (max savings) | **€41** | **€42** |

**Max heat:** for each site, raise availability as high as possible while keeping LCOH below that country's gas heat cost. Goal: **serve the most heat that still beats gas**.
**Max savings:** for each site, pick the availability that maximizes total € savings vs gas:
served heat × (gas cost − LCOH). Goal: **largest bill savings**, even if that means serving less heat.

## Methods




We model an off-grid solar with heat pump and li-ion battery for <200°C and with industrial heat battery for >=200°C processes. Both can be drop-in technology for steam generation in manufacturing facilities.

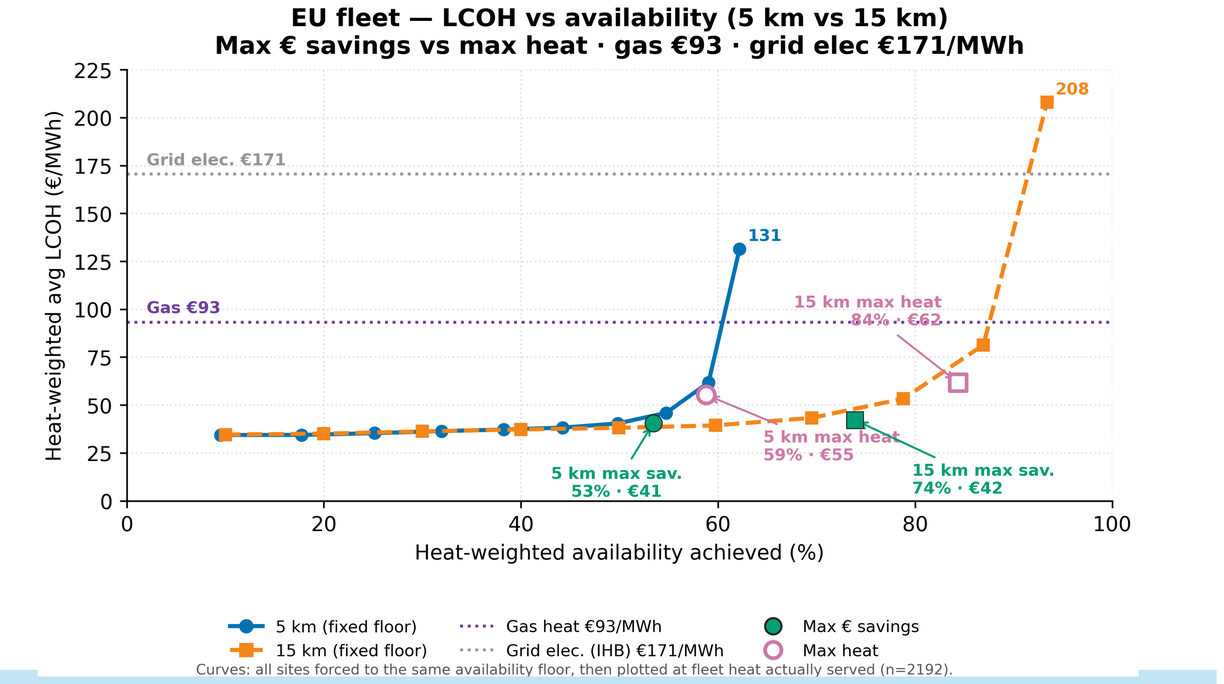
**Facility-level Production Data** | **Solar Radiation** | **Land Availability**

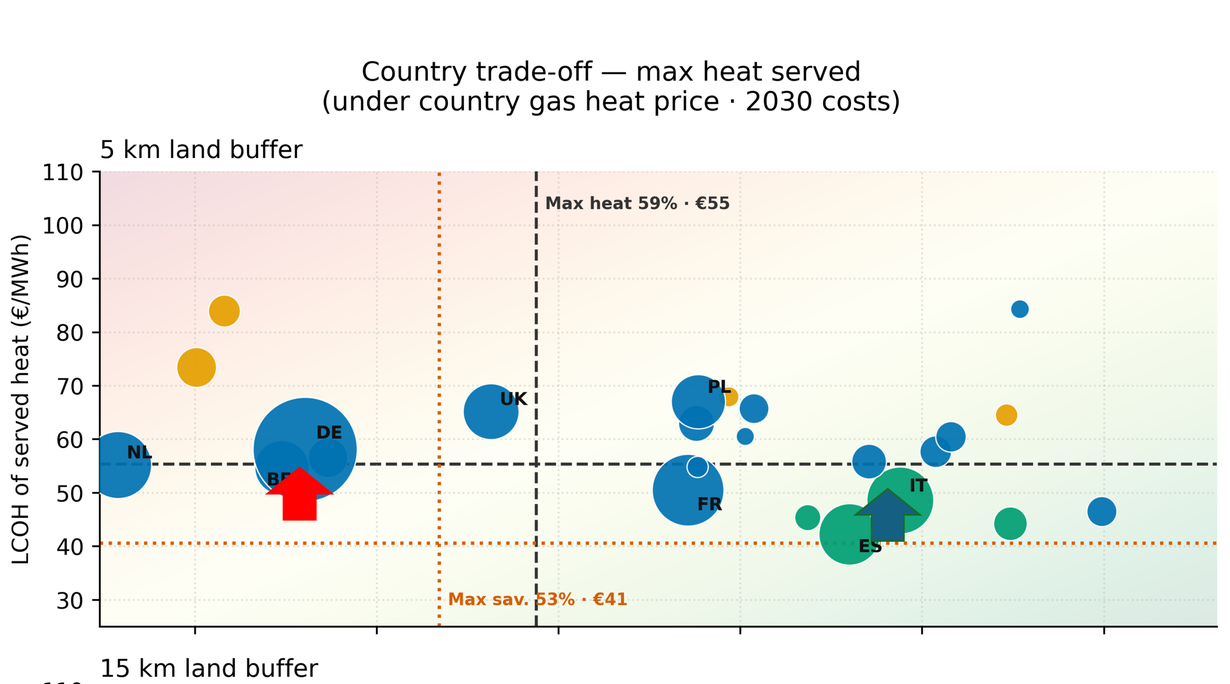
**Optimization**
Minimize LCOH subject to energy balance, solar capacity and land availability

**% of total heat decarbonized and savings**

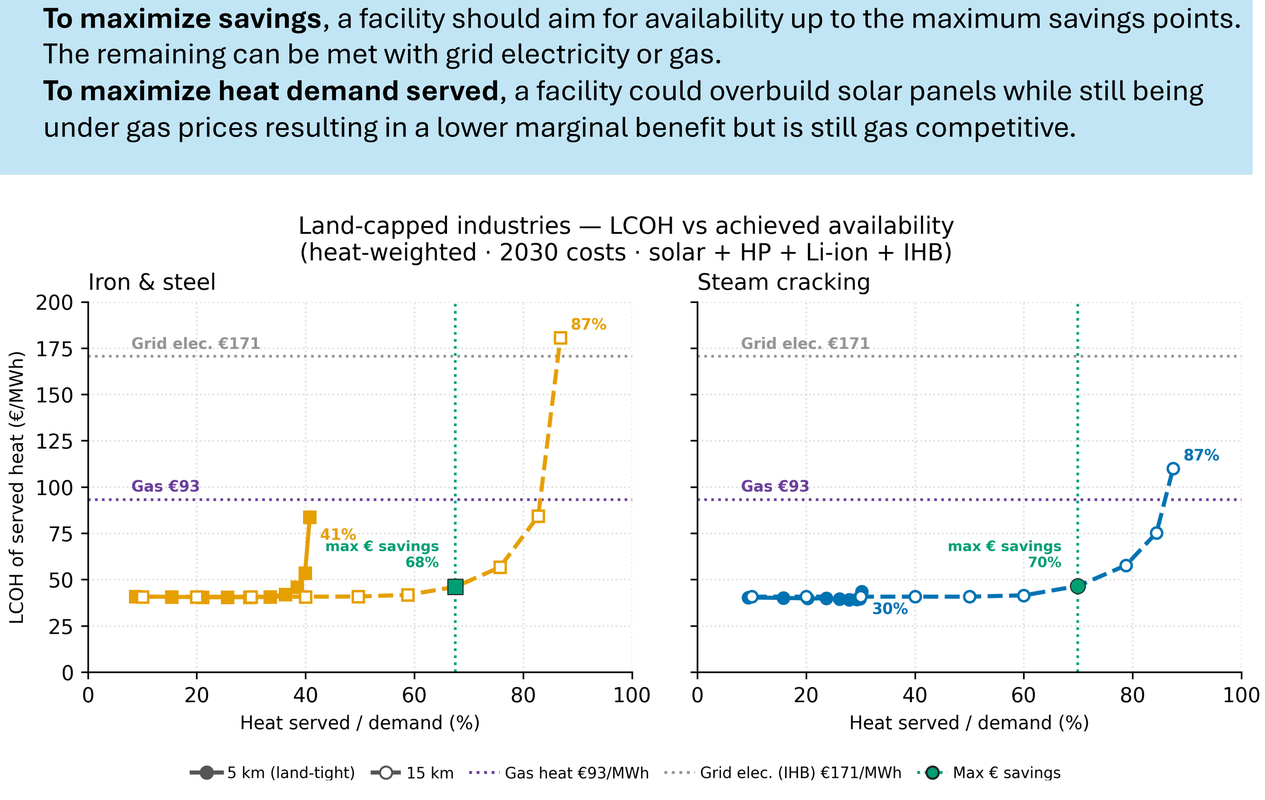
With the total heat level decarbonized at 10-100% availability, we analyzed the costs and benefit of implementing solutions at different availability served for different sectors and countries.

## Discussion




**To maximize savings**, a facility should aim for availability up to the maximum savings points. The remaining can be met with grid electricity or gas.
**To maximize heat demand served**, a facility could overbuild solar panels while still being under gas prices resulting in a lower marginal benefit but is still gas competitive.




- Steam cracking, iron & steel sector have very large sites and have more heat intense processes ( > 500°C) resulting in much higher heat loads.
- These sectors require more hybrid solutions: increased land availability, utilizing grid electricity, co-firing with gas plants.
- Even building up to the max savings point, solar plus storage can meet over 50% of the heat served.

## Conclusion

- Germany, the Netherlands, and France have very large process-heat demand. With a ~15 km land buffer, a large majority can be served below country gas. At 5 km, Germany and the Netherlands remain more land-constrained.
- Southern Europe (Greece, Spain, Italy, Portugal) is the strongest fit: ~95% of demand served at LCOH roughly **70–75% of country gas** (closer to ~half gas under a max-savings framing). Sites tend to be smaller, with better solar yield.
- For high-temperature processes, this solution can decarbonize a high share of **replaceable** heat (~80% for steel and steam cracking at 15 km), even if not 100% everywhere.
- By meeting heat on-site with solar + storage, this package can reduce reliance on gas boilers and incremental electric load on the local grid, potentially easing upgrades and gas exposure, and offers a deployable path to cheaper cleaner heat.